

Definition of the Derivative

Date_____ Period_____

Use the definition of the derivative to find the derivative of each function with respect to x .

1) $y = -2x + 5$

2) $f(x) = -4x - 2$

3) $y = 4x^2 + 1$

4) $f(x) = -3x^2 + 4$

5) $y = -4x^2 - 5x - 2$

6) $y = 3x^2 + 3x + 3$

7) $y = \sqrt{-3x - 5}$

8) $f(x) = \sqrt{4x - 5}$

9) $y = \frac{1}{x + 2}$

10) $f(x) = -\frac{2}{2x - 1}$

Critical thinking question:11) Use the definition of the derivative to show that $f'(0)$ does not exist where $f(x) = |x|$.